

Sankarshan Kulkarni

✉ ksankarshan1208@gmail.com |  [Sankarshan03](#) |  [sankarshan-kulkarni](#)

PROFESSIONAL SUMMARY

Quantitative researcher with deep expertise in **options pricing**, **volatility surface modelling**, **systematic backtesting**, and **risk analytics**. Proficient in **Python**, **SQL**, and **statistical methods**. Proven track record of building high-performance algorithmic trading systems and risk frameworks for multi-billion dollar portfolios.

WORK EXPERIENCE

Associate (Y3) – PricewaterhouseCoopers Services LLP, India Mar 2026 – Present

Bengaluru, Karnataka

- Built quantitative risk analytics and forecasting frameworks for **\$10B+** AUM hedge fund portfolios across Equities, Fixed Income, MBS, and Alternative Investments.
- Developed **Python/SQL** models for risk forecasting and portfolio analytics across **100K+** instruments, reducing manual effort by **1000+ hours annually**.

Technology Analyst (C10) – Citibank (CSIPL) Jul 2024 – Feb 2026

Pune, Maharashtra

- Designed scalable **Python**, **Spark**, and **SQL** data pipelines for quantitative reporting and predictive analytics on terabyte-scale financial datasets.
- Engineered analytical infrastructure and feature engineering for **100+** datasets, improving data quality and research productivity.

KEY PROJECTS

BankNifty Dispersion Trade Monitor  [GitHub](#)

- Developed a real-time dispersion calculator tracking net premium by buying BankNifty ATM straddles and selling constituent ATM straddles, normalised to a **Rs. 6 crore** reference portfolio.
- Built with **Flask**, **Zerodha KiteTicker WebSocket**, and multi-threaded API calls; achieved **75% reduction** in API calls via smart caching with automatic fallback.

BankNifty Short Strangle Backtesting  [GitHub](#)

- Built a high-performance 1-minute intraday backtest engine processing **10M+** rows of options data with entry at 09:20 (strikes near Rs. 50) and exit at 15:20 or **50% stop loss**.
- Computed **Sharpe ratio**, **CAGR**, **max drawdown**, skewness (-1.36), and kurtosis (0.37); optimised runtime **45 seconds**.

SVI Volatility Surface Modelling (SPX)  [GitHub](#)

- Extracted OTM SPXW options (0/1/2 DTE) from a **7.5M-row** dataset; derived forward price ($r=4.5\%$) and computed **Black-76 implied volatilities**.
- Fitted a **5-parameter SVI** model via least squares (**RMSE 0.00256**); validated no butterfly arbitrage via call price convexity.

Kalman Filter Pair Trading  [GitHub](#)

- Developed a dynamic mean-reversion statistical arbitrage strategy on US equities using **Kalman Filter** state-space modelling for rolling hedge ratio estimation.
- Achieved **Sharpe 1.20**, **Calmar 2.0**, max drawdown **20%**, and positive net returns after transaction costs and slippage.

Risk Parity & Minimum Volatility Portfolio

 [GitHub](#)

- Implemented risk parity via marginal risk contribution and min-vol with product constraint on six US stocks.
- Achieved daily variance **10× lower** than market-cap weighted benchmark; risk parity weights evenly distributed (0.12–0.20).

Optimised Intraday EMA-RSI Trading Strategy

 [GitHub](#)

- Engineered a **Numba-JIT** accelerated backtesting engine (**5–10x speedup**) with multi-timeframe EMA(3/10/50) + RSI(14) and pending entry logic.
- Built an interactive **Streamlit** dashboard with real-time performance metrics, drawdown analysis, and Sharpe tracking.

Black-Scholes PDE – Finite Difference (Heat Equation)

 [GitHub](#)

- Transformed the Black-Scholes PDE to the heat equation and solved it numerically using an **explicit finite difference** scheme on a discretised log-moneyness domain.
- Validated against the analytical Black-Scholes formula, achieving **$R^2 = 0.995$** ; quantified numerical convergence and stability conditions.

EDUCATION

2020 – 2024 B.Tech, Chemical Engineering, **Indian Institute of Technology Gandhinagar**

TECHNICAL SKILLS

Languages & Tools	Python, SQL , C++, Matlab, Spark, Pandas, NumPy, SciPy, Statsmodels , Docker, Git, Linux, Streamlit, Numba, Flask, WebSockets
Quantitative Domains	Factor Research, Portfolio Construction, Risk Modeling, Statistical Arbitrage, Time Series Analysis, Monte Carlo Simulation, Derivatives Pricing (Black-Scholes, Black-76), CAPM, Risk Parity, SVI Volatility Modelling , Backtesting, Implied Volatility, Finite Difference Methods

ACHIEVEMENTS

- Global Rank **87** among **4000+** participants in NK Securities Quant Research Challenge for volatility surface construction and option mispricing identification.
- Winner – Algorithmic Trading Hackathon among **2000+** participants for designing systematic quantitative trading strategies and robust statistical backtesting frameworks.